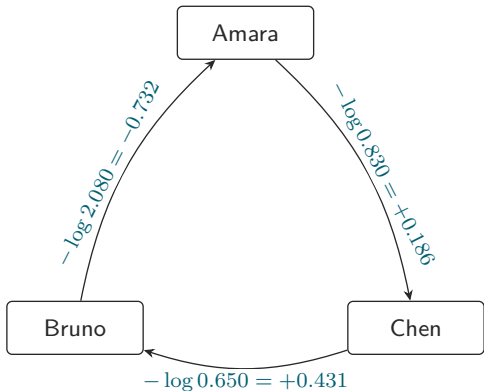




weights $w = -\log(\text{rate})$; $\prod \text{rate} > 1 \iff \sum w < 0$
a profitable loop is a negative cycle; Bellman–Ford finds it in sorted order, deterministically